

Yingrui Hou

Biostatistics & Data Science | Modelling, ML Validation & Decision Support

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SUMMARY

Statistical data scientist with a PhD in experimental physics and several years of experience developing reusable Python/C++ workflows for probabilistic modelling, machine-learning validation, time-dependent inference, and decision support on large, noisy datasets.

Experienced in extracting latent information when ground truth is incomplete, aligning mismatched datasets, testing model robustness, quantifying uncertainty, and translating analytical results into calibration or software decisions.

Seeking to transfer this quantitative and collaborative R&D experience to breeding innovation while developing domain expertise in quantitative genetics and plant-science data.

CORE SKILLS

- Statistics: probabilistic modelling, maximum-likelihood estimation, mixture models, joint fitting, uncertainty quantification
- Data science and ML: Python, NumPy, pandas, scikit-learn, statsmodels, CatBoost / Gradient Boosting
- Analytical methods: supervised learning, feature engineering, time-series modelling, domain adaptation, reweighting
- Validation: train/test diagnostics, model validation, calibration, bias correction, robustness studies, monitoring
- Software and reproducibility: C/C++, Linux, Git, CI/CD, YAML / Shell / CMake automation, shared codebases
- Visualisation and scientific computing: matplotlib, seaborn, ROOT / RooFit, Geant4, large-scale data workflows

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher | CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration | 2021-2025

- Developed reusable Python/C++/ROOT analytical workflows for weak-signal extraction, time-dependent inference, and model validation on large datasets with incomplete event-level ground truth.
- Implemented probabilistic and maximum-likelihood models, including joint fits, to estimate latent parameters under sample mismatch, finite resolution, and biased observations.
- Built CatBoost-based classification workflows with feature studies, train/test diagnostics, threshold scans, and reproducible model application.
- Applied gradient-boosting reweighting and control-sample calibration to align datasets before downstream prediction and statistical inference.
- Contributed reusable fit components, covariance-aware checks, pseudoexperiments, and uncertainty-stress tests for advanced inference workflows.
- Built monitoring and diagnostic workflows that converted expected-versus-observed discrepancies into calibration follow-up and software corrective actions.
- Worked in Linux-based shared codebases with Git, review discipline, reproducible workflows, and international multidisciplinary teams.

Doctoral Research | University of Chinese Academy of Sciences | 2016-2021

- Developed simulation-backed statistical analyses combining model fitting, signal reconstruction, calibration, and bias-aware interpretation of noisy measurements.

SELECTED OUTCOMES

- Improved reconstructed-energy resolution by about 2x through time-resolved signal analysis, component fitting, and calibration on realistic measurement data.
- Validated a statistical extraction strategy through control samples, dataset-specific corrections, and explicit robustness checks under background, efficiency, and resolution effects.

SELECTED PROJECTS

Advanced Inference Workflow Validation

Contributed the proof-of-principle validation needed to move a new basis-expansion inference idea from theory toward a usable workflow under realistic measurement conditions.

Methods: basis expansion, covariance-aware joint fitting, pseudoexperiment validation, statistical background subtraction

Domain Adaptation for ML

Built an ML validation and sample-alignment workflow in a shared analysis environment for noisy selection problems under dataset shift.

Methods: CatBoost training, train-test diagnostics, gradient-boosting reweighting

Time-Series Inference

Built a bias-corrected time-series inference workflow for data affected by selection effects, detector resolution, and sample mismatch.

Methods: acceptance modelling, resolution calibration, custom fit components, joint fitting

Weak-Signal Inference

Built a reusable C++/Python/ROOT workflow for weak-signal extraction in data where background dominates and event-level truth is unavailable.

Methods: likelihood fitting, control-sample validation, production yield workflows

Test-Data Validation & Root-Cause Analysis

Built a production-style validation workflow for early Run 3 photon reconstruction, focused on finding actionable detector and software issues.

Methods: benchmark-channel checks, MC-data comparison, detector-level diagnostics

EDUCATION

- Joint Master and PhD in Particle Physics - University of Chinese Academy of Sciences, 2016-2021
- BSc in Applied Physics - China University of Mining and Technology, 2010-2014
- Languages: Chinese: Native speaker, English: C1, French: B1 to B2